

A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 10

### Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• Sigmoidal (S-shaped) growth curve</li> <li>• Slow start (few breeding individuals), then rapid / exponential growth (years 4-10) as abundant food and space mean birth rate far exceeds death rate</li> <li>• Growth slows as density-dependent factors (intra-species competition for food, disease) raise the death rate</li> <li>• Population stabilises / fluctuates around about 920-950 - the carrying capacity - where birth rate equals death rate</li> </ul> | 4     |
| 01.2     | <ul style="list-style-type: none"> <li>• Density-dependent: effect on survival increases with population density - food shortage / intra-species competition, disease spread, predation</li> <li>• Density-independent: chance of survival unaffected by density - drought, flood, volcanic eruption, severe winter</li> </ul>                                                                                                                                                                                 | 2     |
| 01.3     | <ul style="list-style-type: none"> <li>• Without wolves (natural predators removed) the population is regulated only by starvation / disease at a high density</li> <li>• At this density deer over-graze, preventing tree regeneration and damaging habitats / other species; culling keeps numbers below the damaging level (as in Scotland)</li> </ul>                                                                                                                                                      | 2     |
| 01.4     | <ul style="list-style-type: none"> <li>• Rising prey numbers give predators more food so predator numbers rise; high predation then reduces the prey population</li> <li>• Food shortage reduces predators, allowing prey to recover - cyclic fluctuations with predators lagging behind prey</li> </ul>                                                                                                                                                                                                       | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 78-81; AQA spec 3.1.3.4.                                                                                                                                                                                                                                                                                                                                                                                          |       |
| 02.1     | <ul style="list-style-type: none"> <li>• Tropical intertidal coastal forests of halophytic (salt-tolerant) trees in saline, oxygen-deficient (anaerobic) mud</li> <li>• Prop / aerial roots for support in soft sediment and for gas exchange (pneumatophores); salt excretion</li> <li>• Seeds germinate on the parent tree and float, rooting where they touch sand - rapid natural regeneration</li> </ul>                                                                                                  | 3     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Marks |
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| 02.2     | <ul style="list-style-type: none"> <li>• Coastal erosion / storm protection: roots and vegetation absorb wave energy (least tsunami damage in 2004 where mangroves were intact)</li> <li>• Fisheries: nursery grounds where dense roots protect young fish that later live on reefs / open sea</li> <li>• Protection of coral reefs: slow river water so suspended solids settle and nutrients are absorbed before reaching reefs</li> <li>• Resources: timber, fuel; medicines (antimicrobials against MRSA, cyanobacterium for lung cancer); high biodiversity (crabs, shrimps, sponges, birds)</li> </ul>                                                                | 4     |
| 02.3     | <ul style="list-style-type: none"> <li>• Clearance for aquaculture (shrimp ponds), urban development, ports, tourism; pollution; sea level rise / climate change; destruction of coral reefs that shelter the coast</li> <li>• Legal protection / designated areas (eg Sri Lanka protected all remaining mangroves in 2015)</li> <li>• Replanting and natural regeneration; control of damaging activities; Ramsar sites</li> </ul>                                                                                                                                                                                                                                         | 3     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 66-67; AQA spec 3.1.2.3.4.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| 03.1     | <ul style="list-style-type: none"> <li>• Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish</li> <li>• Species occupy different niches / trophic levels so they do not compete and all food sources are used</li> <li>• Bottom feeders re-suspend nutrients boosting plankton / plant growth</li> <li>• Predators control over-breeding species (tilapia) so remaining fish grow larger; total yield exceeds any single species</li> </ul>                                                                                                          | 4     |
| 03.2     | <ul style="list-style-type: none"> <li>• IMTA: fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton)</li> <li>• Wastes of one species become nutrients for another so less pollution and extra harvests</li> <li>• Aquaponics: aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress)</li> <li>• Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish); rice-fish paddies similar</li> </ul> | 4     |
| 03.3     | <ul style="list-style-type: none"> <li>• Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages</li> <li>• Production still depends on (and removes juveniles from) the wild population; carnivores need large fish inputs</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                 | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |
| 04.1     | <ul style="list-style-type: none"> <li>• Extensive cereal 5.0; intensive cereal 3.0</li> <li>• Intensive beef 0.1</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2     |
| 04.2     | <ul style="list-style-type: none"> <li>• Intensive cereal is most productive per hectare (60 GJ) but less efficient (3.0) than extensive (5.0)</li> <li>• Each extra unit of input gives a smaller extra yield (law of diminishing returns)</li> <li>• Beef has very low productivity and efficiency because energy is lost at an extra trophic level</li> </ul>                                                                                                                                                                                                                                                                                                            | 3     |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
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| 04.3     | <ul style="list-style-type: none"> <li>Any input that increases productivity but requires the use of energy (usually fossil fuel)</li> <li>Manufacture of nitrate fertiliser (Haber process) and pesticides; fuel for machinery; pumping irrigation water; grain drying; food processing and transport; embodied energy of machinery</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 3     |
| 04.4     | <ul style="list-style-type: none"> <li>Because of diminishing returns, the first units of input on unfertilised land give the largest yield increases</li> <li>Moving inputs from intensively farmed land (where extra input adds little) to extensively farmed land gives a greater total yield - though land, cost and distribution limit this</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-327; AQA spec 3.5.1.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| 05.1     | <ul style="list-style-type: none"> <li>Photosynthesis and aerobic respiration balance O<sub>2</sub> and CO<sub>2</sub> in a dynamic equilibrium; evapotranspiration adds water vapour forming clouds and rain; microbes drive the nitrogen and other biogeochemical cycles so wastes do not accumulate and nutrients are recycled</li> <li>Detritivores, fungi and bacteria decompose dead organic matter releasing nutrients and forming humus that binds soil; organic acids weather rock forming new soil; roots bind soil, reducing erosion and regulating water flow / flooding</li> <li>Species provide food, pollination (bees, moths, birds, bats - more reliable than wind, especially in forests), seed dispersal (animals carry larger seeds to suitable sites, faeces as fertiliser, gut passage stimulates germination) and habitat provision (trees as nest sites and canopy microclimate; mollusc shells for hermit crabs)</li> </ul> | 6     |
| 05.2     | <ul style="list-style-type: none"> <li>Its nectar is at the end of a 30 cm tube reached only by the long tongue of the sphinx moth, which carries pollen only between orchids of the same species</li> <li>Advantage: no pollen wasted on other species and the moth has an exclusive food source; risk: if either species declines the other cannot survive</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2     |
| 05.3     | <ul style="list-style-type: none"> <li>They are not ruminants so seeds are not destroyed by repeated chewing / pass through intact; they eat the fruit of most tree species</li> <li>Not territorial so they carry seeds over large distances / areas</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 18-19; AQA spec 3.1.2.1.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| 06.1     | <ul style="list-style-type: none"> <li>Images / maps of large areas; land-use change, deforestation, flooding</li> <li>Vegetation density and surface temperature</li> <li>Wind velocity, sea-surface altitude, wave height, oil pollution</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 3     |
| 06.2     | <ul style="list-style-type: none"> <li>Microwaves pass through clouds (unlike visible light)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1     |
| 06.3     | <ul style="list-style-type: none"> <li>Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected</li> <li>No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
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| <b>06.4</b> | <p><i>Any 4 from:</i></p> <ul style="list-style-type: none"> <li>• Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night</li> <li>• Each tiger has a unique pattern of facial / body stripes</li> <li>• Images are compared with the database so individuals are recognised rather than counted twice</li> <li>• The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings</li> <li>• Limitation: only works for species with unique visible features</li> </ul>                                          | 4     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |
| <b>07.1</b> | <p><i>Any 2 from:</i></p> <ul style="list-style-type: none"> <li>• Hedgerows</li> <li>• Hay meadows</li> <li>• Grazed moorland / heathland</li> <li>• Chalk grassland</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
| <b>07.2</b> | <ul style="list-style-type: none"> <li>• Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive</li> <li>• Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        | 2     |
| <b>07.3</b> | <ul style="list-style-type: none"> <li>• The introduced (non-indigenous) predator may eat other species as well as the pest</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |       |
| <b>08.1</b> | <ul style="list-style-type: none"> <li>• Negative feedback responds to any change by re-establishing the previous equilibrium</li> <li>• Only large unusual events overwhelm the feedback so a new equilibrium forms</li> <li>• Example: methane hydrate release at the end of the Permian causing climate change and mass extinction</li> </ul>                                                                                                                                                                                                                                                                                                                          | 3     |
| <b>08.2</b> | <ul style="list-style-type: none"> <li>• In a diverse system each component is relatively less important so the loss / change of one has a small effect; alternative food sources and relationships buffer the change</li> <li>• Natural: coral reefs and tropical rainforests with complex food webs are stable; simple polar / desert systems fluctuate</li> <li>• Human: agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather</li> <li>• Diverse technical systems: a society using a range of renewable energy resources is more secure than one relying on a single source</li> </ul> | 4     |
| <b>08.3</b> | <p><i>Any 3 from:</i></p> <ul style="list-style-type: none"> <li>• Melting permafrost releasing methane</li> <li>• Reduced albedo as ice / snow melt</li> <li>• Methane hydrate release</li> <li>• Forest and peat fires</li> <li>• Faster soil decomposition</li> <li>• Ocean acidification reducing sequestration</li> <li>• Cirrus cloud formation</li> </ul>                                                                                                                                                                                                                                                                                                          | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
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| <b>09.1</b> | <ul style="list-style-type: none"> <li>• Demersal trawl: seabed species - cod, haddock, plaice, shrimps, scampi; chains / weights damage the seabed (mix aerobic and anaerobic layers, destroy sea fans, corals, seagrass); high by-catch in mixed fisheries; high fuel use</li> <li>• Purse seine: shoaling pelagic species - tuna, sardines, herring, anchovy; can remove whole shoals; dolphins associated with tuna shoals may be caught</li> <li>• Pelagic long-line: tuna, squid; albatrosses (long-lived, low breeding rate) drown on baited hooks; sharks and turtles caught</li> </ul> | 6     |
| <b>09.2</b> | <ul style="list-style-type: none"> <li>• Shrimp trawls use very small mesh so crabs, molluscs, sea urchins, starfish, juvenile fish cannot escape - by-catch can be 99% of the catch</li> <li>• Drift nets are non-selective curtains near the surface catching whales, dolphins, turtles, sharks</li> </ul>                                                                                                                                                                                                                                                                                    | 2     |
| <b>09.3</b> | <ul style="list-style-type: none"> <li>• Lost or discarded nets and traps continue to catch and kill organisms; dead animals act as bait attracting more</li> <li>• Biodegradable ropes / panels so lost traps fall apart; radio transmitters on traps so they can be recovered; net recovery schemes</li> </ul>                                                                                                                                                                                                                                                                                | 2     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 340-343; AQA spec 3.5.2.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
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| 10.1     | <p><b>Levels of response</b></p> <p><b>Level 3 (7-9 marks):</b> Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.</p> <p><b>Level 2 (4-6 marks):</b> Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.</p> <p><b>Level 1 (1-3 marks):</b> Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (credit other relevant, accurate points)</p> <ul style="list-style-type: none"> <li>• Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions</li> <li>• Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection</li> <li>• Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE</li> <li>• Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution</li> <li>• Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels</li> <li>• Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population</li> <li>• Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons</li> <li>• Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support</li> </ul> | 9     |
| 10.2     | <ul style="list-style-type: none"> <li>• Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
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| 11.1     | <p><b>Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Importance: resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation)</li> <li>• Causes: agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY; trends by region</li> <li>• Consequences: loss of resources, biodiversity and species, fragmentation; hydrological changes; soil impacts; climate impacts (albedo, carbon, rainfall)</li> <li>• Productivity vs biodiversity: non-indigenous species, monocultures, close planting, simple age structure, clear-felling</li> <li>• Sustainable management: MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China); Rio Bravo case study</li> <li>• Evaluation: economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures; judgement</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.</p> | 25    |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
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| 11.2     | <p><b>Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.</b></p> <p><b>Levels of response</b></p> <p><b>Level 5 (21-25 marks):</b> Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.</p> <p><b>Level 4 (16-20 marks):</b> Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.</p> <p><b>Level 3 (11-15 marks):</b> Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.</p> <p><b>Level 2 (6-10 marks):</b> Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.</p> <p><b>Level 1 (1-5 marks):</b> Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.</p> <p><b>0 marks:</b> No relevant content.</p> <p><b>Indicative content</b> (students are not expected to cover all of these; credit other relevant material)</p> <ul style="list-style-type: none"> <li>• Temperate broadleaf woodland: seasonality, deep soils, high biodiversity, resources, climate and soil regulation; deforestation, fragmentation, management change; SSSIs, ancient woodland protection, coppicing, mixed planting</li> <li>• Tropical rainforest: stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils; fuelwood, timber, agriculture, mining, reservoirs, climate change; protected areas, debt-for-nature swaps, sustainable exploitation</li> <li>• Tropical coral reefs: cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity; fisheries, erosion protection, medicines, tourism; damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species; MPAs, no-take zones, control of activities</li> <li>• Deep-water reefs: cold, dark, slow growth; research, fisheries; trawling, oil / gas, acidification; SACs</li> <li>• Oceanic islands: isolation, endemism, no mammal predators; exploitation, introduced species, habitat change, sea level; eradication, control of development and visitors</li> <li>• Mangroves: halophytes, anaerobic mud; erosion protection, fisheries nurseries, sediment trapping; clearance for aquaculture / urban / ports, pollution, climate change; protection, replanting</li> <li>• Antarctica: extreme cold, seasonality, marine nutrients, ice; water store, albedo, carbon, research; climate change, ozone, tourism, fishing, minerals; Antarctic Treaty, CCAMLR, tourism and waste rules</li> <li>• Synthesis: features -&gt; importance -&gt; threats -&gt; strategy; evaluation of success; conclusion</li> </ul> <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4.</p> | 25    |